Gemma 4 Is Less Self-Biased Than It Looks

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github.com/Vein05/gemma4-paper-track Supplement: `rubber-stamp-supplement.pdf` Kaggle paper track

Abstract

Coding agents often decide whether a patch is finished by asking a model, often the one that wrote it. Studies of self-preference test whether a judge is biased toward its own outputs by comparing its scores on them with its scores on other models' outputs. The closest study reports that the preference shrinks but remains against writers as strong as the judge. We ask Gemma 4 31B and eight other models to score SWE-bench patches, and we grade every patch with the hidden tests. By the usual measure, Gemma trusts its own failing patches far more than other writers' failing patches. Against seven writers as strong as Gemma, about half of that gap remains, and Gemma gives their failing patches the top score as often as its own. The larger problem is the patches themselves. Most of Gemma's own failing patches are wrong fixes that pass every existing test. Existing tests catch few of these wrong fixes, and when Gemma is shown an existing test that asserts the behavior the issue changes, it obeys the test and breaks the correct patch. We release every agent and judge trace on GitHub and Hugging Face.

1 Introduction

In matplotlib issue 24570, Gemma 4 31B wrote a patch that fixes the reported alignment bug but also swaps the left and right cases of a shared helper. When we later asked Gemma whether this patch would pass the project's hidden tests, it gave the highest score on our 0 to 100 scale and called the patch "a simple swap" that "correctly implements the expected behavior". GLM-4.5 scored the same patch 20 because it "breaks VPacker by reversing left and right", and the hidden tests for left and right alignment fail. Coding agents make this call at the end of every task, and many make it by asking the model that wrote the patch.

The standard test of this risk compares a judge's scores on its own outputs with its scores on other models' outputs. Pombal et al. (2026) ran this test on contest programs with executed labels. Most judges favored their own programs, and the preference was larger when the judge recognized its own output. Gemma 3 12B and 27B were among the exceptions. But errors from strong writers are harder for a verifier to detect (Zhou et al., 2025), so a judge compared with weaker writers will seem to favor itself even if it never recognizes its own work. Pombal et al. matched writers on ability and found that the preference "attenuates in places but never disappears".

We ran this test on Gemma 4 31B with SWE-bench patches, against two groups of public agents. Against five writers of mixed strength, the usual kind of comparison, Gemma scores its own failing patches 16.3 points further above judges from other model families than it scores the other writers' failing patches. Against the seven writers that solve about as many issues as Gemma, the gap is 8.7 points. Gemma gives these seven writers' failing patches the top score of 100 as often as its own, and the weakest writer's a quarter as often.

About half of what looks like self-preference is thus trust in any equally strong writer's failing patch. Most of Gemma's own failing patches are wrong fixes: the issue's tests fail while every other existing test passes. Neither the checks that Gemma runs itself nor the existing tests catch most of them. The existing tests do catch regressions, and Gemma repairs many of these when shown the broken test. When a test asserts the old behavior that the issue asks to change, however, Gemma obeys the test and undoes a correct fix.

We make the following contributions:

- **Gemma 4's self-preference halves against equally strong writers.** Gemma's extra trust in its own failing patches is 16.3 points against five writers of mixed strength and 8.7 against seven of its own strength. It gives the seven's failing patches the top score as often as its own (Section 3.1).
- **Gemma gives the top score to most strong writers' failing patches in Python, Go, Ruby and PHP.** Most of its own are wrong fixes, and the checks we ran through Gemma missed most of them (Section 3.2).
- **Existing tests catch regressions, but outdated tests mislead the repair.** Shown the existing tests its patch breaks, Gemma repairs real regressions. On Lite it breaks 12 of 14 correct patches held back by a test that asserts the old behavior. On other writers' patches, having Gemma label outdated tests first spares most correct patches the repair would break (Section 3.3).

2 Setup

Patches. The main data set is 193 held-out issues, a random sample of SWE-bench Verified issues (Jimenez et al., 2023) outside SWE-bench Verified Mini (MariusHobbhahn). We call the model that wrote a patch its writer. The 4-bit competition checkpoint of Gemma 4 31B made one attempt per issue in mini-SWE-agent v1.7.0 (SWE-agent) at temperature 0, and we graded its patches with SWE-bench's grading code. Other writers' patches are public mini-SWE-agent submissions (SWE-bench experiments) with their official labels. We compare Gemma with two groups of them. Five writers of mixed strength stand for the usual comparison: GLM-4.5, gpt-oss-120b, Kimi-K2, Llama-4-Maverick and Qwen3-Coder. The seven equally strong writers are every public writer whose share of issues solved is within 8 points of Gemma's: GLM-4.5 and Kimi-K2 again, o3, o4-mini, GPT-5 mini, GPT-5 nano and Gemini 2.5 Pro. We removed test files from every patch and dropped very long patches, leaving 180 from Gemma.

Two more data sets test whether the result depends on our run or our language. Another team ran Gemma 4 31B on the 300 SWE-bench Lite issues, in mini-SWE-agent 2.2.1 at full precision with thinking on. We ran Gemma on 120 Go, Ruby and PHP issues from SWE-bench Multilingual (SWE-bench) in mini-SWE-agent 2.0.0 with thinking on. There the other writers are four recent models, GLM-5, Kimi-K2.5, DeepSeek-V3.2 and MiniMax-M2.5.

Judges. Nine models scored every held-out patch: Gemma 4 31B, gpt-oss-20b, Qwen3.5-9B, GLM-4.7-Flash and the five mixed-strength writers. For the five writers that are only in the second group, they scored the failing patches. Each judge saw the issue and one patch, without its writer, and scored the chance that the patch passes the hidden tests from 0 to 100. Calls went through OpenRouter at temperature 0 with reasoning off (gpt-oss at low effort). Fewer judges scored the Lite and Go, Ruby and PHP patches, chosen before any score was read.

Measures. A judge's score on a failing patch says little alone, because some failing patches look right to everyone. We therefore measure how far a judge scores each patch above the mean of the outside judges, every judge from outside its own model family and the writer's. Extra trust is how much further above them a judge scores one group of failing patches than another group. We report 95% intervals from 2,000 bootstrap resamples of issues.

SWE-bench grades a patch with two groups of tests. The issue's tests fail before the reference fix and pass after it. The other existing tests must keep passing. We call a failing patch a wrong fix when the issue's tests fail and every other test passes, and a regression when it breaks one of the other tests.

Plans. We committed a written plan, with the rule for reading its result, before computing each test. The rule that picks the equally strong writers was fixed before their patches were scored. The split by failure type was chosen after we saw results, and we mark it as exploratory.

3 Results

3.1 Gemma's self-preference halves against equally strong writers

Against the mixed-strength writers, Gemma's extra trust in its own failing patches is 16.3 points, and about the same on Lite (Figure 1b). Weak writers supply much of this gap. Gemma gives 100 to most of its own failing patches and to a quarter or less of the two weakest writers', and these two supply most of the group's failing patches (Figure 1a).

Against the equally strong writers, Gemma gives 100 to 59% of its own failing patches and to 57% of theirs, a difference within noise (Table 1). Its extra trust in its own failing patches is 8.7 points, about half of the gap above, with a 95% interval above zero. Four of the seven are OpenAI models, so both comparisons give each of the four model families (OpenAI, Google, Moonshot, Zhipu) equal weight.

Table 1. Gemma 4 31B's scores on failing held-out patches, its own and the seven equally strong writers'. "Above outside judges" is Gemma's mean score minus the outside judges' mean on the same patches, in points.

Writer	Issues solved	Gemma scores 100	Above outside judges
o3	62%	55%	−1.7
GLM-4.5	61%	57%	+3.8
GPT-5 mini	59%	79%	+3.0
Kimi-K2	59%	55%	+6.4
Gemini 2.5 Pro	55%	65%	+12.6
Gemma 4 31B	54%	59%	+13.3
GPT-5 nano	48%	26%	−12.1
o4-mini	48%	41%	−7.4

The size of the remaining gap depends on which writers are compared. Gemma scores Gemini 2.5 Pro's failing patches as far above the outside judges as its own, and the four OpenAI writers' near or below them. Without Gemini the gap is larger, and without the OpenAI writers its interval includes zero. Gemini comes from Gemma's maker, but one writer cannot show whether that matters. We also could not tell whether the remaining gap follows the content of Gemma's patches or Gemma recognizing them (Supplement, Section D).

3.2 Gemma trusts strong writers' failing patches, and its own are mostly wrong fixes

Gemma trusts stronger writers' failing patches in Go, Ruby and PHP too. There every other writer solves more issues than Gemma. Gemma gives 100 to 76% of its own failing patches and to 68% of theirs, and its extra trust in its own is within noise. The outside judges give Gemma's failing patches 100 only 29% of the time.

What are these patches? Most of Gemma's own are wrong fixes. We sorted Gemma's 82 failing held-out patches by failure type (exploratory; Supplement, Section C). 52 are wrong fixes, in which every existing test still passes. The other 30 break an existing test.

Checks that Gemma runs itself miss most of the same failures (Table 2; Supplement, Section E). Gemma's thinking mode did not help, although Chen et al. (2025) found that long chains of thought before evaluation reduce harmful self-preference.

Table 2. Checks run by Gemma, and what happens to its failing patches.

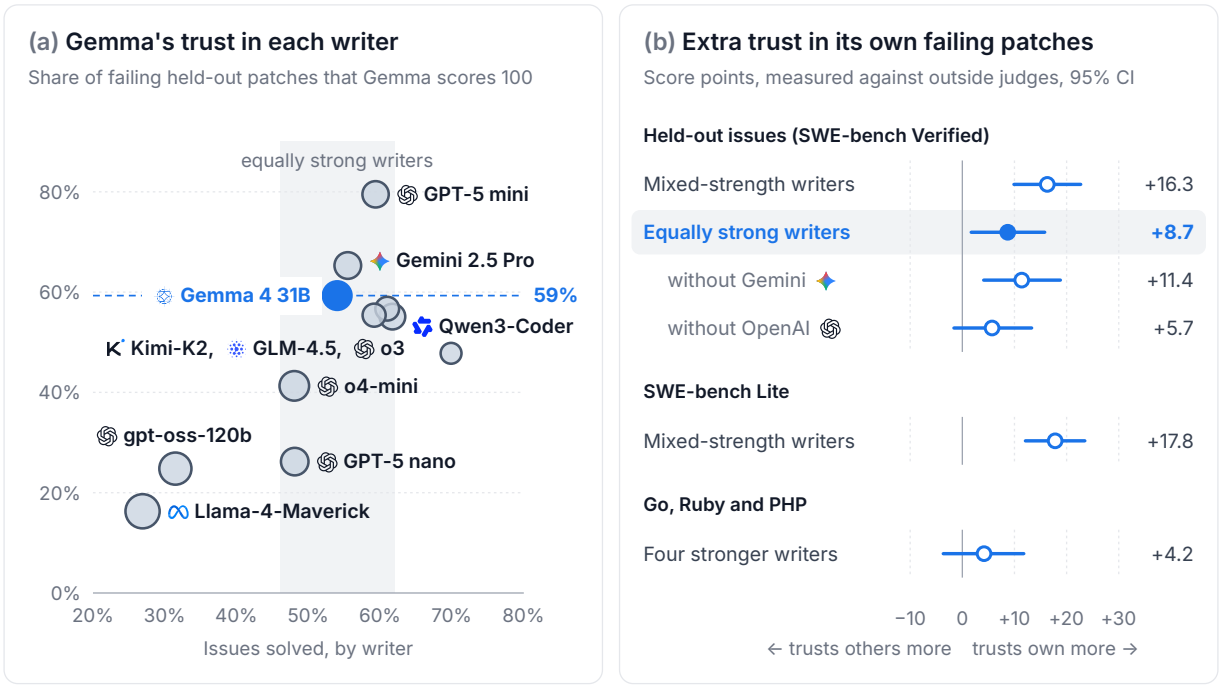


Figure 1. Gemma 4 31B trusts failing patches more when their writer is stronger. (a) Held-out issues. The shaded band holds the equally strong writers, and dot area shows the number of failing patches. (b) Gemma's extra trust in its own failing patches, with 95% intervals, halves against the equally strong writers.

Check run by Gemma	What happens to Gemma's failing patches
Thinking before scoring (Verified Mini issues)	still scores 62% of its failing patches 100, against 61% without thinking
Its own reproduction script, written while fixing the issue	the last run ends without an error in almost all failing runs that ran one
A shell with the patch applied; Gemma must write a test that fails without it	on the 49 failing held-out patches it had scored 100, still scores 79% at 100
Fine-tuning Gemma (LoRA) on labeled patches from other issues	still accepts almost all failing patches it had scored 100, on new held-out and Lite issues

3.3 Existing tests catch regressions, but outdated tests mislead the repair

A wrong fix passes every test on SWE-bench's list, so a check built on existing tests mostly reaches regressions. Our check runs the existing tests, on that list or not, that match the changed code before and after the patch, and holds back the patch if a test that passed before now fails. A held-back patch gets one more Gemma agent run. In one condition Gemma sees the broken tests and is asked to make them pass and keep the issue solved without editing the test files. In the other, the second look, Gemma is only told that the patch may have mistakes (Supplement, Section F).

The check holds back 22 of Gemma's 30 failing held-out patches that break a listed test, and only 4 of its 52 wrong fixes. Shown the broken tests, Gemma repairs 14 of the 26 held-back failing patches, against 7 with a second look (Table 3). On Gemma's Lite patches, the same repair fixed 1 of 30. Half of those 30 were wrong fixes. Their broken tests are outside SWE-bench's list, so making them pass cannot fix the issue.

Table 3. The repair on two sets of Gemma's patches, with the broken tests shown and with a second look. A

held-back patch “breaks an existing test” when it fails one of SWE-bench’s listed regression tests. Repaired counts leave out one Lite issue whose test fails at random.

	Held-out, tests	Held-out, second look	Lite, tests	Lite, second look
Held-back failing patches	26	26	30	30
Of these, break an existing test	22		15	
Repaired	14	7	1	0
Held-back correct patches	9	9	14	14
Broken by the repair	4	1	12	2
Net change in passing patches	+10	+6	−11	−2

The larger cost falls on correct patches. For all 14 correct Lite patches that the check held back, the reference fix fails the same check, so the test asserts the old behavior. Gemma broke 12 of these 14 patches when shown such a test, against 2 with a second look. On xarray issue 5131 it wrote that “the constraints of the task (making tests pass without editing them) force me to restore it”. On held-out issues the repair still helps on net: 108 of Gemma’s 180 patches pass after it, against 98 before. On Lite it hurts.

The missing step is a decision about which tests may change, which Agentless (Xia et al., 2024) leaves to the model. We tested one rule on the mixed-strength writers’ held-back patches. Gemma first labels each broken test as outdated or a real regression, and the repair ignores the outdated ones. Correct patches broken fell from 20 to 7, but fewer failing patches were repaired, and the net gain is within noise, which our plan counts as failure.

4 Related work

Measuring self-preference. Panickssery et al. (2024) link a judge’s preference for its own summaries to how well it recognizes them. Pombal et al. (2026) asked judges whether a LiveCodeBench program passes each unit test. Most judges wrongly pass their own programs more often than other families’ programs. Roytburg et al. (2026) compare a judge’s own wrong answers with other models’ wrong answers to the same question, and much of the measured self-preference goes away. Their correction concerns the judge. Pombal et al. also match the other writers on ability, and the preference shrinks but remains. Kozman et al. (2026) report both effects: a model that wrote an answer “approves its own errors 11.9 percentage points more often”, and stronger generators’ errors are harder to catch. We apply the same match to Gemma 4 and repository patches and find the same.

Verifying strong writers. On math, knowledge and reasoning tasks, Zhou et al. (2025) find that weak generators’ errors are easier to detect than strong ones’. Figure 1a shows the same pattern for repository patches. Khullar et al. (2026) find that one model rates a SWE-bench patch as more correct when it is framed as its own earlier answer, a cue our judges never see.

Tests as a check. In test-based program repair, test suites are often too weak, so repair tools produce many incorrect patches that pass them (Xiong et al., 2017). Our wrong fixes are this class. When code changes, some tests must change too, and Shang et al. (2026) find that coding agents identify those tests with an F1 below 50%. ImpossibleBench (Zhong et al., 2025) edits SWE-bench tests to conflict with the issue and tells the agent to put the issue first. Our conflicts come from the repositories’ own tests, and our note asked for both without ranking them.

5 Limitations

The gap that remains against equally strong writers depends on which writers are compared. The failure-type split is exploratory and covers Gemma's patches only. The Go, Ruby and PHP run also changes the agent version, thinking and the language. In Gemma's held-out agent runs, the repository's Python environment was not active, so Gemma often could not import the code. This lowers its resolve rate and may produce more crash patches, which the repair fixes most easily. The repair's harm rests on 23 correct patches, and each Lite repair condition ran once.

6 Conclusion

A judge that seems to favor its own patches may partly be favoring strong writers. For Gemma 4 31B, about half of the measured preference for its own failing patches remains once the comparison writers are as strong as Gemma, and Gemma gives their failing patches the top score as often as its own. For an agent, the risk is Gemma's trust in wrong fixes, which neither its own checks nor the existing tests catch. An agent built on Gemma should not let Gemma's score decide that a patch is finished. When it adds the existing tests, it also needs a rule for which tests the issue may change.

7 Data, code and compute

The GitHub repository holds the paper, supplement, code, plans and judge answers, and the Hugging Face dataset holds the agent runs and prompts. Running `./reproduce.sh main` regenerates every table in this paper from the released data, without a GPU. The project used about 113 GPU-hours. At RunPod on-demand prices this compute costs about $191.

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